A-A-50734
15 May 1988

COMMERCIAL ITEM DESCRIPTION

MULTIMETER

The General Services Administration has authorized the use of this Commercial Item Description.

Scope. This Commercial Item Description (CID) defines a ruggedized, autoranging, 3200 count, digital/analog multimeter used for measuring alternating current (AC), direct current (DC), resistance, conductance (optional). AC voltage, DC voltage, continunity (audible tone), and diode (or transistor junction) voltage drop. The multimeter is intended for use in flight line maintenance applications.

Applicable documents.

Specifications, Standards and Handbooks. The following specifications, standards, and handbooks form a part of this CID to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specification and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment, General Specification for

STANDARDS

MILITARY

MIL-STD-461	Electromagnetic Emission and Susceptability Requirements for the Control of Electromagnetic Interference.

(Copies of specifications, standards, handbooks and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

AMSC N/A FSC 6625

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Order of precedence. In the event of conflict between the text of this specification and the references cited herein (except for the associated detail specifications, specification sheets for MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Salient characteristics.

1. General specification. The multimeter specified by this CID shall comply with the type II, class 2, style A, color R requirements of MIL-T-28800. When the two specifications conflict, this commercial item description shall govern.

2. Safety. The multimeter shall comply with type I safety requirements specified in MIL-T-28800.

a. Electrical shock hazard. The multimeter shall be designed to reduce electrical shock hazards to using personnel. The only metal exposed on the outside of the body and panel which may be exposed on the inside are the input connectors.

b. Input connectors. To reduce shock hazard, all multimeter input connectors shall be recessed banana jacks and the test lead input connectors shall be shrouded banana plugs. The input connectors shall not touch the IEC Rigid Test Finger as defined in IEC-65 and as referenced in MIL-T-28800.

c. Accessories. All accessories shall be suitable for use or storage at -40 degrees C. (i.e. will not become rigid or brittle)

3. Parts, materials, and processes. Unless otherwise specified herein, the multimeter shall comply with type III parts, materials, and processes specified in MIL-T-28800.

a. Restricted materials. The multimeter shall comply with the restricted material requirements of MIL-T-28800. In addition, materials or combination of materials shall not be used which, through outgassing or other physical phenomena, cause deterioration of other materials or degradation of performance in the multimeter or adjacent systems.

4. Design and construction. Unless otherwise specified herein, the multimeter shall comply with the type III design and construction requirements in MIL-T-28800. The multimeter shall use solid-state devices. Electron tubes shall not be used. All circuits shall be constructed on plug-in subassemblies and printed circuit cards wherever possible.

5. Electrical power requirements. The only power from which the multimeter shall operate is the power supplied by the battery contained within the multimeter.

a. Battery. One 9V battery, National Electronic Distributors Association (NEDA) type 1604 9V alkaline, or equal, shall be used within the multimeter. In voltage or current measurement functions (under worse case battery drain conditions), the battery shall be capable of sustaining 300 hours of operation at 25°C, and 150 hours of operation at -20°C or 55°C. The battery may be shipped installed.

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b. Battery test. A display indication shall be provided to show marginal performance of the multimeter due to low battery potential.

6. Weight. Less than 4 pounds with cover and all accessories.

a. Dimensions. Less than 4" H x 6" W x 13" L with cover on.

7. Enclosure. The multimeter shall comply with the style A enclosure requirements of MIL-T-28800; except that the enclosure, handle, latches and any identification labels shall be non-metallic and non-conductive. The handle may be rigid.

8. Marking and identification. Marking and identification shall be in accordance with the type III requirements of MIL-T-28800; except the identification plate shall be non-conductive with durability equivalent to the front panel markings and may be placed on the rear panel.

a. Panel markings. Panel markings shall be in accordance with type I of MIL-T-28800.

9. Environmental requirements. Except as specified in paragraph 3.9 and 3.15, the multimeter shall comply with the Class 2 requirements of MIL-T-28800; including loose cargo bounce, high impact shock, and fungus resistance.

a. Electromagnetic interface (EMI). The applicable EMI requirements of MIL-T-28800 shall be as specified in (a) and (b) below. Except as specified in (b) the full requirements of MIL-STD-461 apply.

(1) RE02

(2) RE03 - one volt/meter from 14kHz to 1 GHz

b. Storage temperature range. -55°C to 85°C.

c. Operating temperature range. -40°C to 55°C. From -40°C to -15°C, response time may be degraded.

d. Humidity and water resistance. The multimeter shall comply with MIL-T-28800 for a Style A, Class 2 instrument.

10. Reliability. The multimeter shall comply with the type III reliability requirements specified in MIL-T-28800.

11. Maintainability. The multimeter shall comply with the type III maintainability requirements specified in MIL-T-28800 modified as follows:

a. The MIL-T-28800 type I maintenance provisions restricting the need for special selection of replacement parts shall apply.

b. The MIL-T-28800 maintenance provisions requiring that any special tools and patch cables required for maintenance and calibration of the multimeterbe provided and stowed within the multimeter (non-access to the user) shall apply.

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c. The certification time shall not be greater than one hour. Certification time includes all time required to verify performance on all range and function settings of the multimeter. Certification time does not include the time to repair and readjust and multimeter as a result of a malfunction.

12. Interchangeability. The multimeter shall comply with the type III interchangeability requirements of MIL-T-28800.

13. Functionally description. The multimeter shall be capable of measuring direct current (DC) voltages from ± 0.1 millivolt (mV) to ± 1000 volts; alternating current (AC) voltages from 0.1 millivolt (mV) to 1000 volts; resistance from 0.1 ohm to 30.00 Megohms (Mohms) directly, and up to 10,000 Megohms (Mohms) using conductance; DC current ± 0.1 microamp (uA) to ± 10.00 Amps (A); AC current from 0.1 microamp (uA) to 10.00 A; diode (or transistor junction) voltage drops to 2.000 volts; and provide an audible tone for continuity measurements below 100 ohms.

14. Input isolation. Any input terminal shall be capable of being floated to at least 1000 volts above earth ground.

15. Accuracy. Unless otherwise specified, all electrical specifications in paragraph 3.15 are from 18°C to 28°C with up to 95% humidity, for one year after calibration. At temperatures less than 18°C or greater than 28°C, the temperature coefficient shall be less than 0.1 x the applicable electrical specification per °C. Accuracy is specified as +/- ([% of reading] + [number of least significant digits]).

a. DC voltage. Ranges, resolution, and accuracy. As specified in Table #1 or better.

TABLE #1	DC VOLTAGE	
RANGE	RESOLUTION	ACCURACY
300.0 mV	0.1 mV	+/-(0.2% + 1)
3.000V	0.001V	"
30.00V	0.01V	"
300.0V	0.1V	"
1000V	1V	"

(1) Input impedance. The input impedance shall be greater than 9.9 Megohms, paralleled by less than 100 picofarads, on any range.

(2) Overload. The multimeter shall be capable of continuously withstanding, without damage or blown fuses, 1000 volts rms on all ranges; except 500 volts rms continuously on the 300.0 mV range.

(3) Normal mode rejection ratio. Normal mode noise rejection shall not be less than 60 decibels (dB) at 50 and 60 Hz (+/-0.15%).

(4) Common mode rejection ratio. Common mode rejection, with a 1000 ohm unbalance in the low lead, shall not be less than 120 decibels (dB) at 50 and 60 Hz (+/- 0.15%).

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a. AC Voltage. Ranges, resolution, and accuracy. As specified in Table #2 or better.

TABLE #2 AC VOLTAGE				
RANGE	RESOLUTION	ACCURACY		
		40 Hz to	2 kHz to 10 kHz	10 kHz to 30 kHz
300.0 mV	0.1 mV	+/-(0.5%+3)	+/-(2%+3)	+/-(4%+10)
3.000V	0.001V	"	"	"
30.00V	0.01V	"	"	"
300.0V	0.1V	"	"	"
1000V	1V	+/-(1%+3)	+/-(3%+3)	Not specified

(1) Input impedance. The input impedance shall be greater than 9.9 Megohms, paralleled by less than 100 picofarads, on any range.

(2) Overload. The multimeter shall be capable of continuously withstanding, without damage or blown fuses, 1000 rms on all ranges; except 500 volts rms continuously on the 300.0 mV range. The multimeter shall be capable of withstanding an AC volt-Hertz product of up to 10.

(3) Common mode rejection ratio. Common mode rejection, with a 1000 ohm unbalance in the low lead, shall not be less than 60 decibels (dB) at 50 and 60 Hz (+ 0.15%).

c. Resistance. Ranges, resolution, and accuracy. As specified in Table #3 or better.

TABLE #3 RESISTANCE		
RANGE	RESOLUTION	ACCURACY
30.00 nanoSiemen	0.01 nS	± (2%+10)
300.0 ohm	0.1 ohm	± (03%+2)
3.000 kilohm	0.001 kohm	± (0.2%+1)
30.00 kilohm	0.01 kohm	"
300.0 kilonm	0.1 kohm	"
3.000 Megonm	0.001 Mohm	"
30.00 Megohm	0.0. Mohm	± (1%+1)

(1) Humidity. The 30.00 nS range accuracy specification shall be limited to 70% maximum relative humidity from 28°C to 55°C.

(2) Maximum current through unknown resistance. The multimeter shall not provide more than 1.1 mA on any range.

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(3) Maximum open terminal voltage. The open terminal voltage shall not be greater than 3.0 volts on any range.

(4) Maximum on scale test voltage. The multimeter shall be capable of allowing in-circuit resistance measurements. It shall not apply greater than 450 mV at full scale on the following ranges: 300.0 ohms, 3.000 kilohms, 30.00 kilohms, 300.0 ohms, and 3.000 Megohms.

(5) Overload. The multimeter shall be capable of continuously withstanding, without damage or blown fuese, 500 volts rms on all ranges.

d. Diode and continuity test.

(1) Continuity test. The multimeter shall provide a continuous audible tone (equivalent) for test resistance below 100 ohms. Forward biased (0.15 volts or greater) diode or transistor junctions shall inhibit continuous tone. Any deviation requires Air Force approval.

(2) Diode test. The multimeter shall be capable of displaying diode or transistor junction voltage drops to at least 2.000 volts. A momentary audible tone shall indicate a forward biased junction, and the absence of sound shall indicate a reversed biased or open junction.

(3) Maximum test current. The maximum diode or continuity test current shall be 1.1 mA.

(4) Maximum open terminal voltage. The open terminal diode or continuity test voltage shall not be greater than 3.5 volts.

(5) Overload. The multimeter shall be capable of continuously withstanding, without damage or blown fuses, 500 volts rms on all ranges.

e. DC current. Ranges, resolution, and burden voltage. As specified in Table #4 or better.

TABLE #4 DC CURRENT

RANGE	RESOLUTION	ACCURACY	MAX BURDEN VOLT.
300.0 uA	0.1 uA	± (0.75%+2)	0.6 mV/uA
3.000 uA	1 uA	"	0.6 mV/uA
30.00 mA	0.01 m	"	7 mV/uA
300.0 mA	0.1 mA	"	7 mV/uA
10.00A	0.01A	"	50 mV/uA

(1) Overload. The multimeter shall be capable of continuously withstanding, without damage, 0.4A rms on all ranges; except the 10.00A range shall be capable of withstanding 10A rms continuously and 20A for 30 seconds. All current inputs shall be fused or protected to interrupt overload currents, up to 10,000A, from accidental voltage overloads in circuits up to 600 volts.

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f. AC current. Ranges, resolution, accuracy, and burden voltage. As specified in Table #5 or better.

TABLE #5	AC CURRENT		
RANGE	RESOLUTION	ACCURACY	MAX BURDEN VOLT.
300.0 uA	0.1 uA	± (1.5%+2)	0.6 mV/uA
3.000 uA	1 uA	"	0.6 mV/uA
30.00 mA	0.01 m	"	7 mV/uA
300.0 mA	0.1 mA	"	7 mV/uA
10.00A	0.01A	"	50 mV/uA

(1) Overload. The multimeter shall be capable of continuously withstanding, without damage, 0.4A rms on all ranges; except the 10.00A range shall be capable of withstanding 10A rms continuously and 20A for 30 seconds. All current inputs shall be fused or protected to interrupt overload currents, up to 10,000A, from accidental voltage overloads in circuits up to 600 volts.

g. Range and function control. All input functions (voltage, current, resistance, and diode/continuity test) shall be controlled with one rotary switch. Ranging shall be automatic (with a manual overide) in DC and AC voltage from the 3.000 V range to the 1000 V range, and in resistance from the 300.0 ohm range to the 30.00 Megohm range. Ranging shall be automatic (with a manual overide) in DC and AC current between 30.00 mA and 300.0 mA; and between 300.0 uA and 3000 uA.

(1) Readout display. The readout display shall include a liquid crystal display (LCD) with a digital readout with 3200 counts of resolution, and analog bar-graph (or equivalent) readout with at least 30 segments of resolution, and auto-zeroing. The digital display shall update at least 2 times per second and the analog display shall update at least 20 times per second.

(2) Automatic polarity. The polarity of the measured value shall be automatically indicated on the digital and analog displays.

(3) Response time. Response time is defined as the time interval between an abrupt change of input and the time and new value is displayed within specified accuracy, including the worst case time to change ranges in the autorange mode.

(a) Direct current or voltage response shall be less than 2 seconds.

(b) Alternating current or voltage response time shall be less than 4 seconds.

(c) Resistance response time shall be less than 2 seconds on all ranges except that in the nanoSiemen range or the lowest and highest resistance ranges, it shall not be more than 5 seconds.

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(4) Maintenance and calibration accessories. If required, the multimeter shall be provided with all necessary maintenance and calibration accessories specified in 3.11(b). Not more than one circuit board extender or patch cable shall be required to permit maintenance of the modules and plug-in circuit cards.

16. Responsibility for inspection. Except for bid sample testing, responsibility for inspection is as specified in MIL-T-28800. Bid sample testing shall be performed by the government.

17. Quality assurance tests. This multimeter may be inspected and tested by bid sample inspection to insure compliance with the quality assurance provisions of MIL-T-28800 and as specified herein.

18. Bid sample inspection. When required by solicitation, procurement of the multimeter shall be by the submission and testing of a bid sample.

19. Number of bid samples. The number of bid samples and operation and maintenance manuals to be submitted shall be required in the solicitation. When required in the solicitation, the offerer shall submit with the bid sample their normal production test procedures and the explosive atmosphere reliability, maintainability and EMI rationale. The Government will, whenever possible, use the offerer's normal production acceptance test procedures to test the bid sample.

a. Explosive atmosphere, reliability, maintainability and EMI rationale. When required by the solicitation, the offeror shall submit clear and concise rationale showing how the multimeter shall comply with the specified explosive atmosphere, reliability, maintainability, and EMI requirements.

(1) Explosive atmosphere rationale. The explosive atmosphere rationale may include data from the previous explosive atmosphere testing on similar equipment or other pertinent data available to the offeror. The rationale shall also include a discussion of design modifications necessary, if any, to meet the requirements of the solicitation. The acceptance of the submitted rationale does not relieve the successful bidder of performing and successfully completing explosive atmosphere tests.

(2) Reliability rationale. The rationale may include data from previous raliability tests, reliability predictions, and other data available to the offeror. In the absence of such data on the multimeter being offered, such data on similar multimeter of equal or greater complexity produced by the offeror may be submitted. However, such data must be clearly identified as comparative data, and accompanied with specifications and technical literature on the similar multimeters. The acceptance of the submitted rationale does not relieve the successful bidder of performing and successfully completing reliability tests.

(3) Maintainability rationale. The maintainability rationale may include data from previous maintainability tests, records of repair and calibration data, and other data on the multimeter available to the offeror. Data on selection of module size and other design features relevant to maintainability may also be submitted. The acceptance of the submitted rationale does not relieve the successful bidder of performing and successfully completing maintainability tests.

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(4) EMI rationale. The EMI rationale may include data from previous EMI testing on similar equipment or other pertinent data available to the offerer. The rationale shall also include a discussion of design modifications necessary if any, to meet the requirements of the solicitation. The acceptance of the submitted rationale does not relieve the successful bidder of performing and successfully completing production EMI tests.

Packaging, packing, and marking. The items shall be packaged in accordance with normal commercial practice and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage. The shipping container shall be in compliance with national motor freight classification and uniform freight classification. Shipping container shall be marked as specified in the contract or order.

Certification. The contractor shall certify that the product offered meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, and standards and quality assurance practices and is the same product offered for sale in the commercial marketplace. Small businesses that cannot offer products with established commercial marketplace acceptability, may compete by submission of bid samples. (Government reserves the right to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract.)

Known acceptable products. The following product(s) are known to have previously met the requirements of this CID and need not to resubmit samples or certification. All other manufacturers must meet the requirements of this CID and submit to bid testing and certification as required.

Manufacturer	Model	NSN
John Fluke Mfg Co	8025B	6625-01-266-3494

Custodian
Air Force - 99

Review activity
Army - CR
Navy - SH
DLA - ES

Preparing activity
Air Force - 82

(Project No. 6625-0693)

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